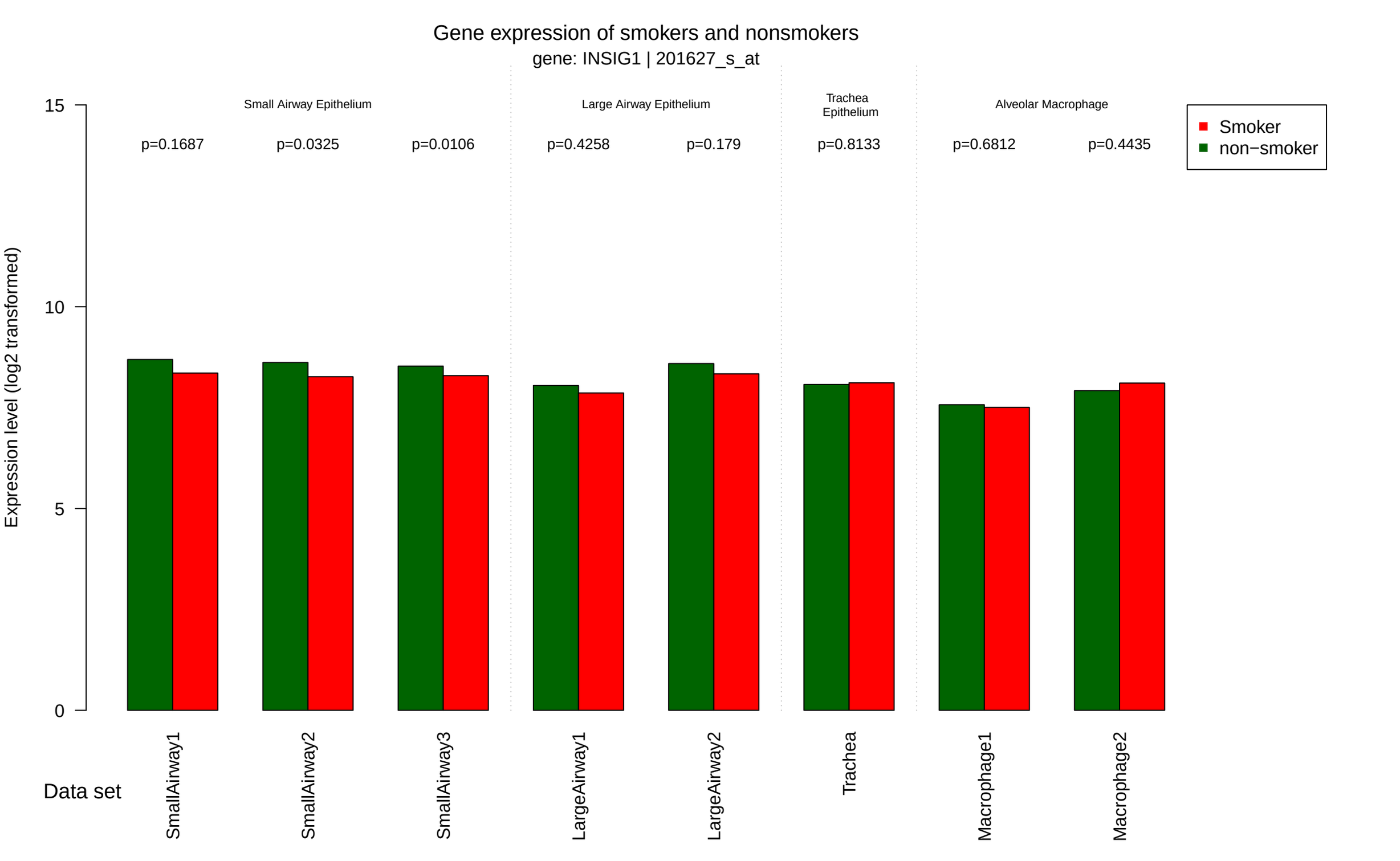
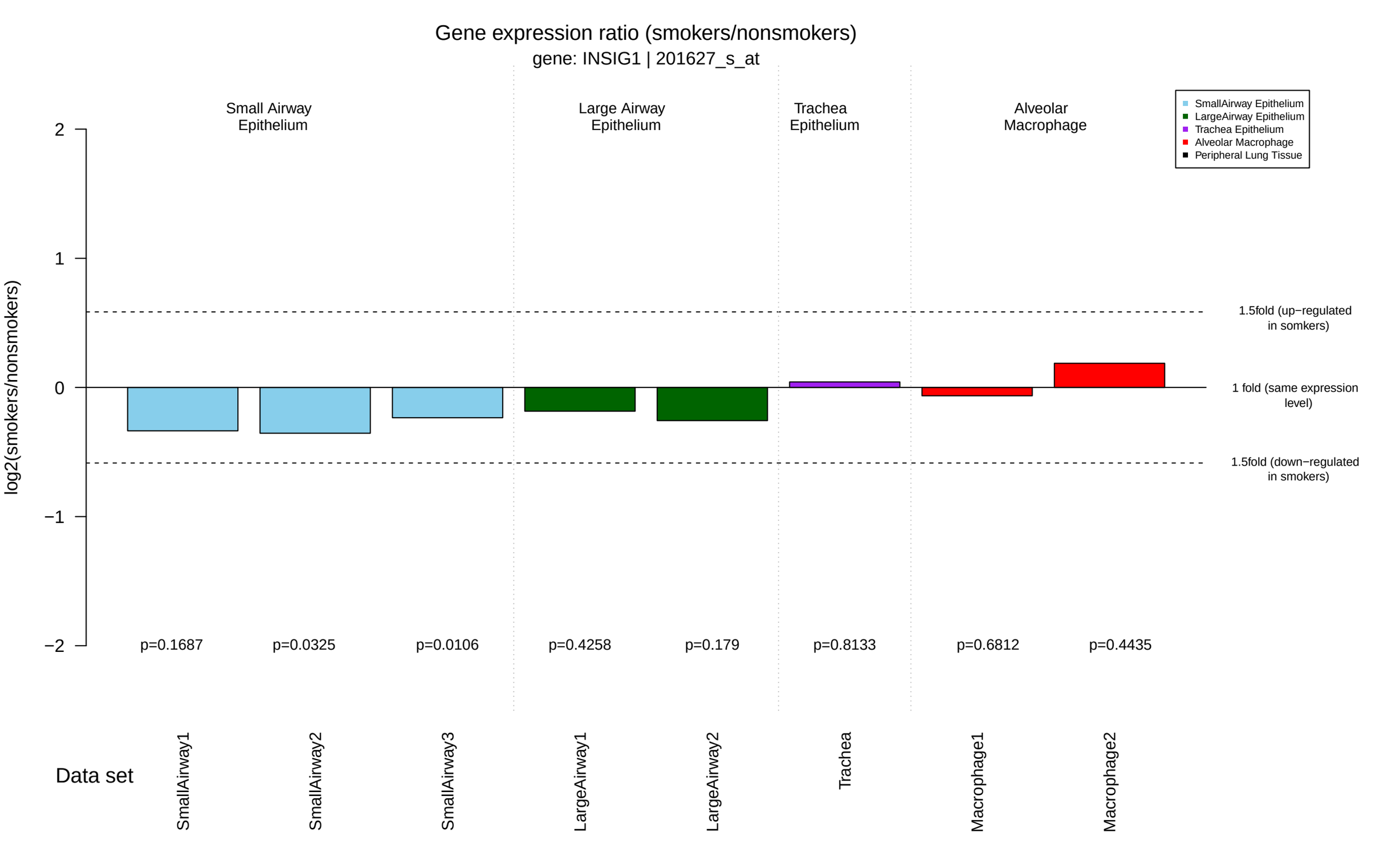
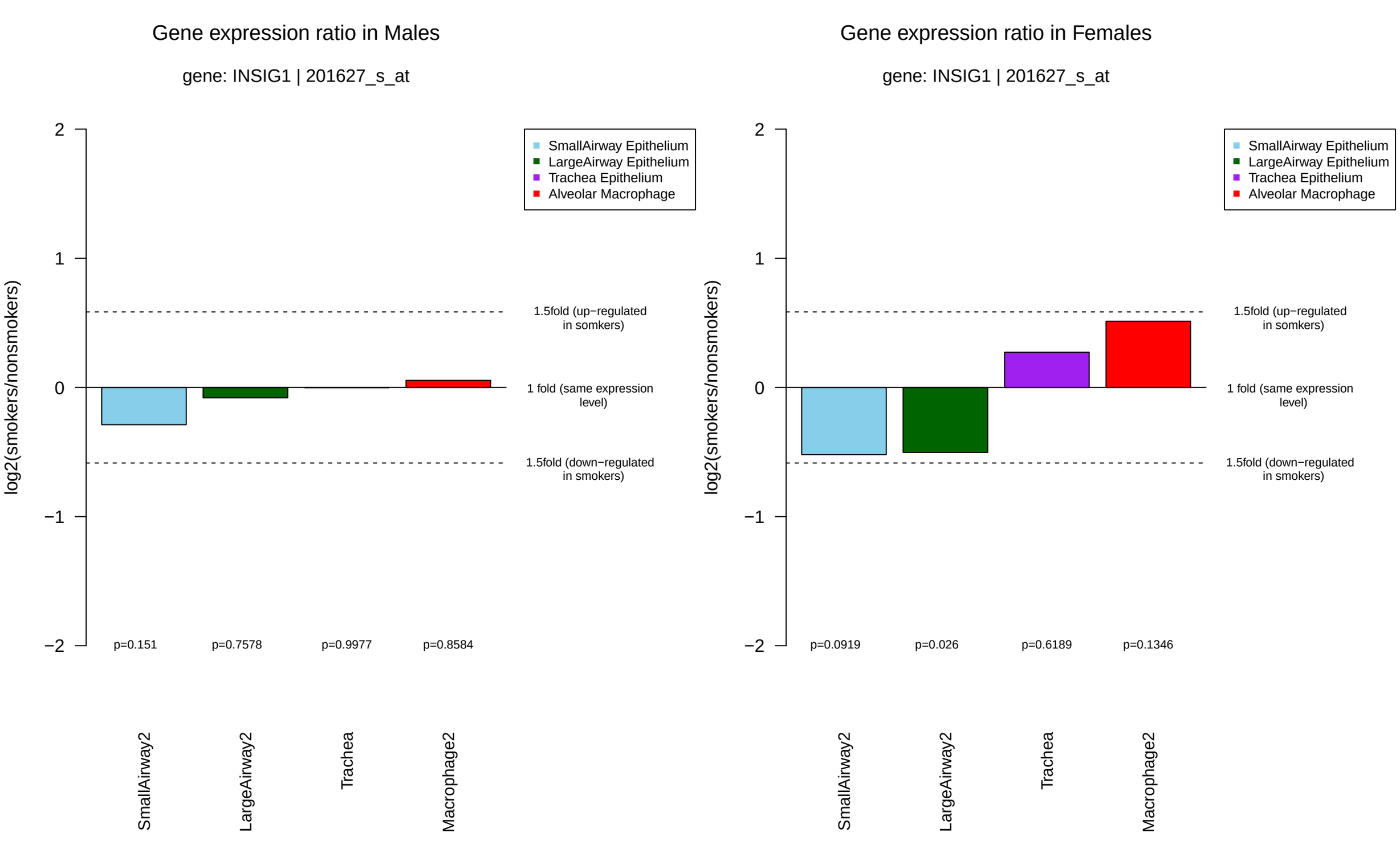
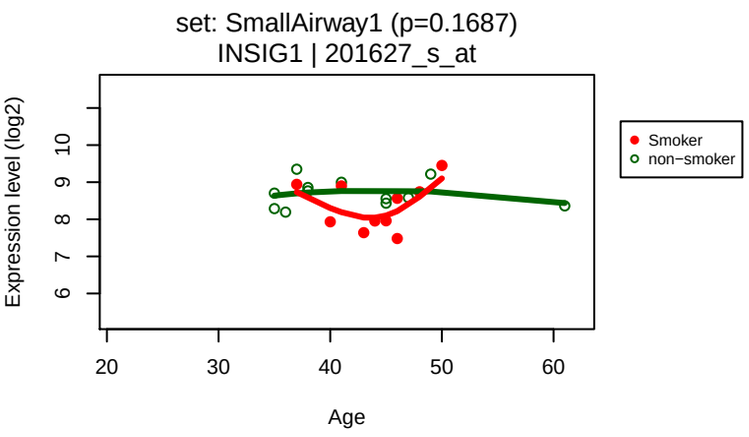
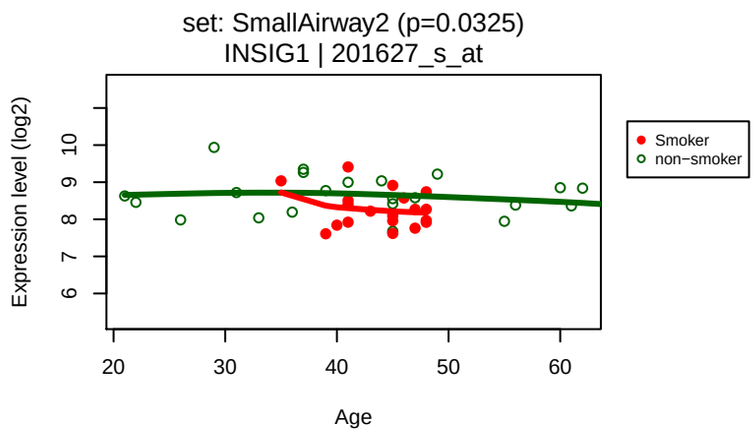
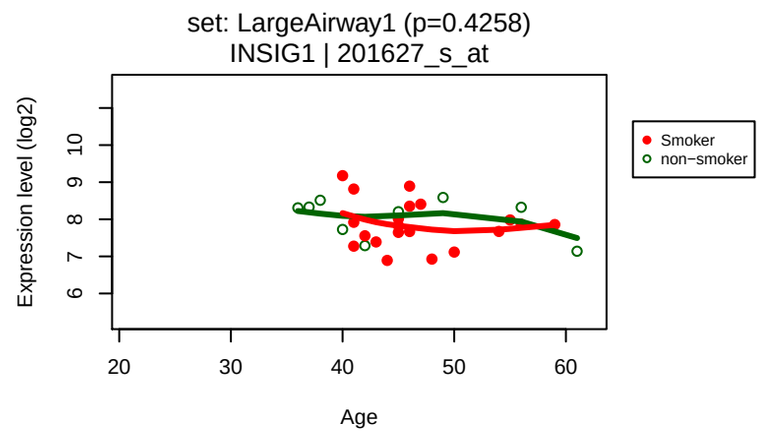

Gene expression of smokers and nonsmokers
gene: INSIG1 | 201627_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1687
p=0.0325
p=0.0106
p=0.4258
p=0.179
p=0.8133
p=0.6812
p=0.4435
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

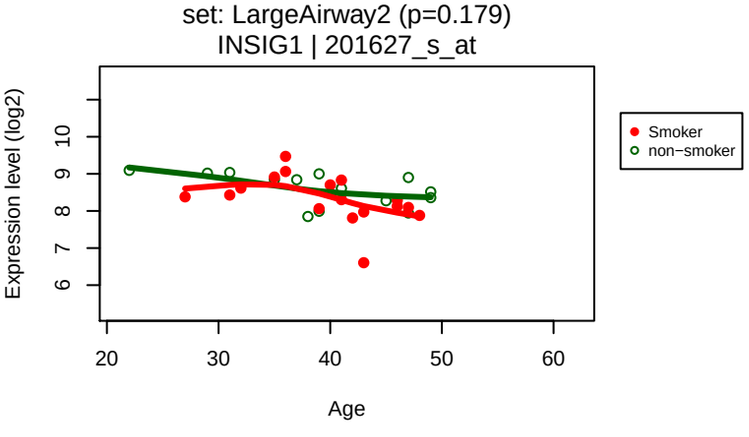

Gene expression ratio (smokers/nonsmokers)
gene: INSIG1 | 201627_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1687
p=0.0325
p=0.0106
p=0.4258
p=0.179
p=0.8133
p=0.6812
p=0.4435
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

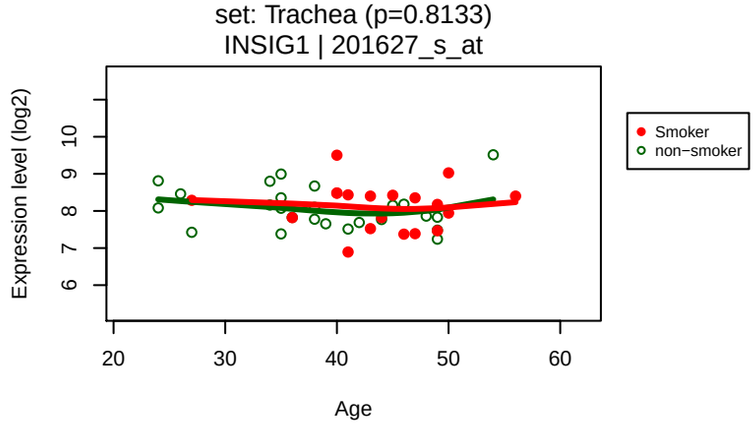

Gene expression ratio in Males
gene: INSIG1 | 201627_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.151
p=0.7578
p=0.9977
p=0.8584
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: INSIG1 | 201627_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0919
p=0.026
p=0.6189
p=0.1346
SmallAirway2
LargeAirway2
Trachea
Macrophage2

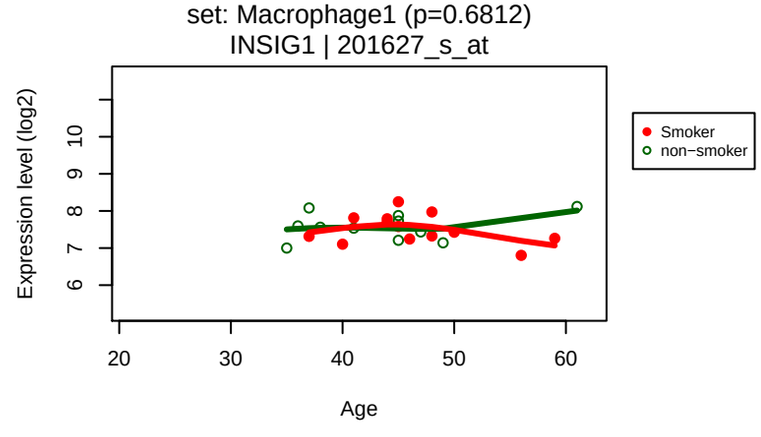

set: SmallAirway1 (p=0.1687)
INSIG1 | 201627_s_at
Expression level (log2)
Age
Smoker
non−smoker

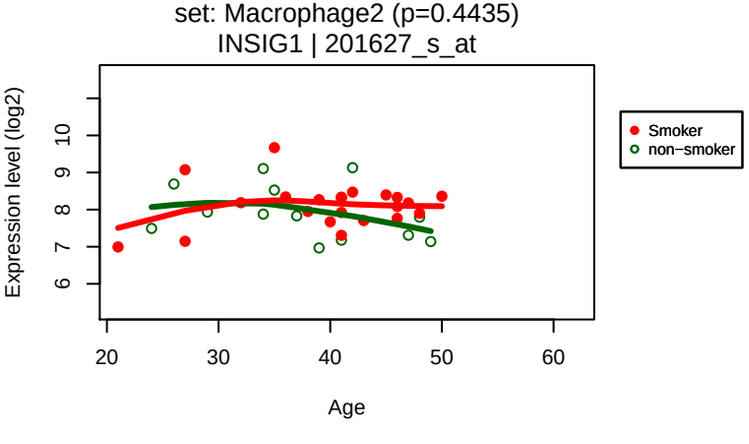

set: SmallAirway2 (p=0.0325)
INSIG1 | 201627_s_at
Expression level (log2)
Age
Smoker
non−smoker

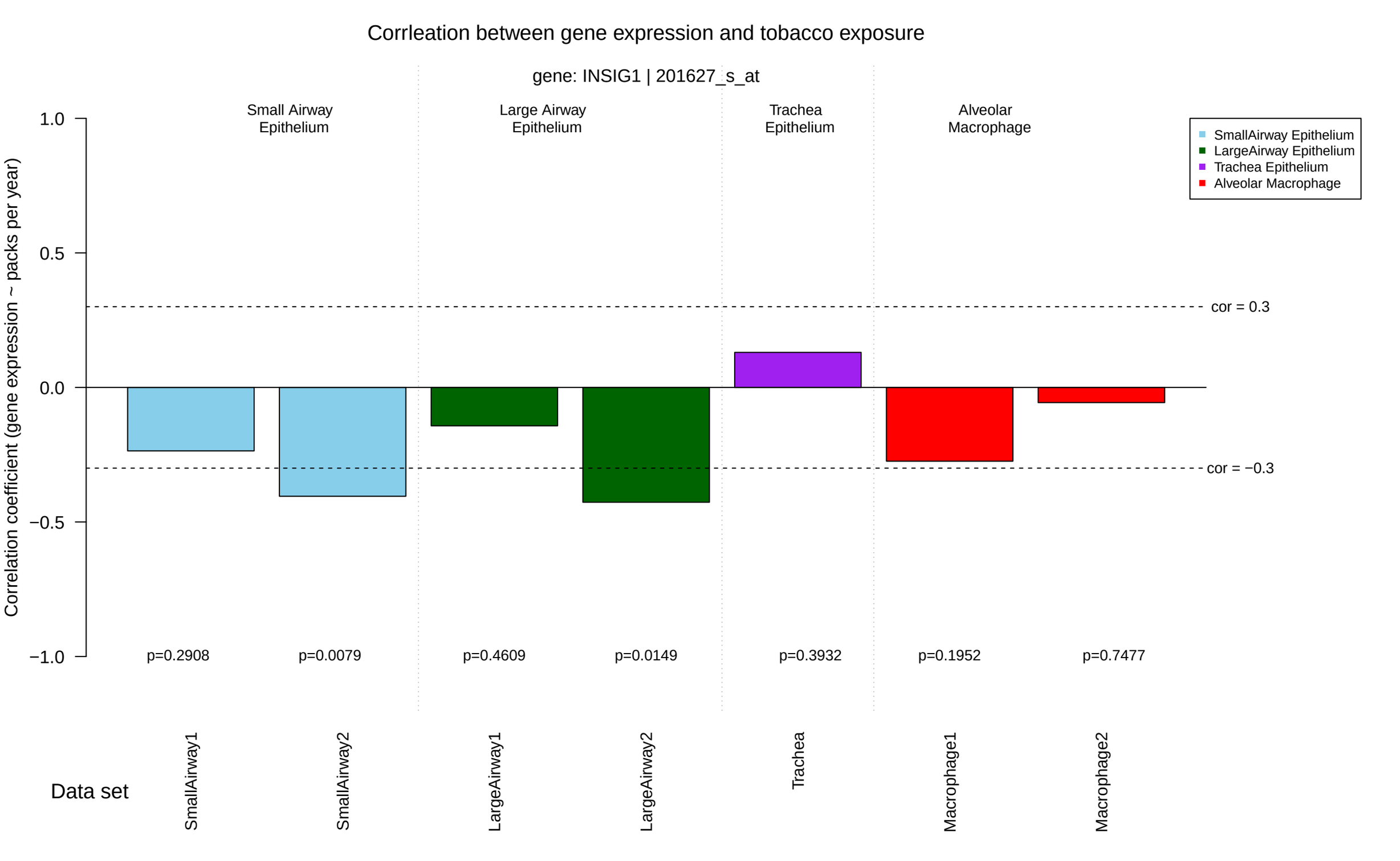

set: LargeAirway1 (p=0.4258)
INSIG1 | 201627_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.179)
INSIG1 | 201627_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8133)
INSIG1 | 201627_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6812)
INSIG1 | 201627_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4435)
INSIG1 | 201627_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: INSIG1 | 201627_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2908
p=0.0079
p=0.4609
p=0.0149
p=0.3932
p=0.1952
p=0.7477
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2